FoodRescue TN — API Contracts

AWD Validation 1 • Proposed, not implemented • Version 2.0

API requests are routed through the Gateway. REST/OpenFeign handles immediate reads and stock reservations; RabbitMQ carries domain events.

01 Donation Offer

Method	Endpoint	Purpose	Responses
POST	/api/offers	Publish surplus offer	201, 400, 403
GET	/api/offers/{id}	Read offer and stock	200, 404
POST	/api/offers/{id}/reservations	Reserve food quantity	201, 400, 409
POST	/api/reservations/{id}/release	Release an unused reservation	200, 404, 409

02 Matching & Pickup

Method	Endpoint	Purpose	Responses
POST	/api/matches	Match offer with charity	201, 404, 409
POST	/api/pickups/{id}/assign	Assign an available driver	200, 404, 409
PATCH	/api/pickups/{id}/status	Update delivery status	200, 403, 404
GET	/api/pickups?driverId=	List driver pickups	200, 401

03 Partner Registry & Trust

Method	Endpoint	Purpose	Responses
POST	/api/partners	Register an organization	201, 400
GET	/api/partners/{id}	Read partner profile	200, 404
PATCH	/api/partners/{id}/verification	Approve or reject organization	200, 403, 404
POST	/api/partners/{id}/reviews	Submit a partner review	201, 400, 403

04 Impact Analytics

Method	Endpoint	Purpose	Responses
GET	/api/impact/summary	Global saved-food totals	200
GET	/api/impact/governorates/{id}	Local impact summary	200, 404
GET	/api/impact/forecast?month=	Predicted demand (planned)	200, 400
GET	/api/impact/reports?period=	Export/report status	200, 400

05 Volunteer Driver

Method	Endpoint	Purpose	Responses
POST	/api/drivers	Create volunteer profile	201, 400, 409
GET	/api/drivers/{id}	Read driver profile	200, 404
PUT	/api/drivers/{id}/availability	Update available time slots	200, 400, 403

Method	Endpoint	Purpose	Responses
GET	/api/drivers/available?lat=&lng=	Find available drivers	200, 400

06 Surprise Basket Marketplace

Method	Endpoint	Purpose	Responses
GET	/api/baskets?governorate=	Browse discounted baskets	200
POST	/api/orders	Reserve and place order	201, 400, 409
GET	/api/orders/{id}	View order and pickup details	200, 403, 404
POST	/api/orders/{id}/collect	Redeem pickup code	200, 404, 409

07 Common User + Notification

Method	Endpoint	Purpose	Responses
GET	/api/users/me	Read application profile	200, 401
PATCH	/api/users/me	Edit profile/preferences	200, 400, 401
GET	/api/notifications	List user notifications	200, 401
PATCH	/api/notifications/{id}/read	Mark notification read	200, 403, 404

Event contracts

Publisher	Event	Key payload	Subscribers
Offer	OfferPublished	eventId, offerId, donorPartnerId	Matching
Matching	MatchConfirmed	eventId, matchId, associationId, driverId	Common notifications
Matching	DeliveryCompleted	eventId, pickupId, foodKg, governorate	Impact Analytics
Marketplace	BasketCollected	eventId, orderId, foodKg	Impact Analytics

Failure and data rules

• 409 Conflict: insufficient stock, repeated order, or invalid state change. 401/403: unauthenticated or forbidden. 404: missing resource. 503: critical dependency unavailable.

• The Offer Service is the only stock owner. Event consumers must ignore duplicate eventId values. Reserve stock synchronously and release after aborted workflows.